Chapter 2

Background

In this chapter we start by reviewing the two main subjects of the thesis which are loudness measurement (Section 2.1) and musical notation (Section 2.2). This will be useful in our understanding of the way we extract information from the audio signal and in our understanding of how this information can be represented as a musical concept, respectively. More details on the interpretation of Chopin's pieces are presented (Section 2.3). We then review existing work in the area (Section 2.4).

2.1 The concept of loudness

"A painter paints pictures on canvas.
But musicians paint their pictures on silence."
—Leopold Stokowski

Loudness is a subjective sensation of the magnitude or strength of sound. Intensity, on the other hand is an objectively measured physical property, an amount of power. It is expressed in power units per unit area, as in watts per square meter. Sound intensity is the major physical determinant of loudness, but the relationship between intensity and loudness is very intricate. The fact that the intensity-loudness relationship is confounded by frequency is especially important for music: Because the ear's sensitivity varies with frequency, equal intensities do not elicit equal loudness across all audible frequencies. The unit for sound intensity is watts per square meter. $1\mathrm{W/m}^2$ is a rather intense sound which approaches the upper limit of hearing. The threshold of hearing at 1000 Hz is around $10^{-12}\mathrm{W/m}^2$.

Sound intensity is usually expressed in terms of intensity level rather than pure intensity. Intensity level implies a comparison of a particular intensity to

some baseline value. The comparisons implied by intensity levels of sound are ratio comparisons which use the unit known as "decibel". Decibels are measures of power ratios and they are best defined in terms of the formula by which they are computed. For sound intensity,

$$D = 10[\log(I/I_0)], \tag{2.1}$$

where D is the number of decibels, I is the particular intensity which the decibel value places on a level, and I_0 is the baseline value. The baseline that is used is the intensity at the threshold of hearing.

It is important to recognise two consequences of the definition of decibels as the logarithm of a ratio. One, a relative small range of decibels encompasses a relatively large range of acoustic powers. Two, one cannot simply combine decibels to obtain an intensity level for tonal combinations. The decibel scale is logarithmic; equal decibel amounts stand for equal ratios.

Sound Pressure Level (SPL) is a more common stimulus measure than intensity level because it is easier to measure and it relates logically to the concept of pressure variations being responsible for sound. It is expressed in decibels which are computed differently in relation to a different baseline. The formula is

$$D = 20[\log(P/P_0)], \tag{2.2}$$

where D is the number of decibels, P is a particular pressure value, and P_0 is a baseline pressure. A common baseline is .00002 newtons per square meter. Most decibel measures in psychoacoustic literature are indicators of SPL.

Volume is the apparent size or extensity of a sound. Volume's existence as an independent tonal attribute is questionable. It has no parallel relationship with any single physical attribute in the way that loudness is related to intensity or pitch is related to frequency.

As referred by [Radocy and Boyle, 2003, pg.52–65], experiments by Stevens [1934] show that an increase in intensity level brought a more rapid volume increase at higher than at lower frequencies. In later research Stevens and Terrace [1962] plotted frequency-intensity combinations used for equal volume judgements and formed a "vol" scale of measurement in which 1 vol equals the apparent volume of 1000 Hz at 40 decibels of sound pressure level. They found that regardless of frequency, at around 140 decibels all tones sounded equal in volume. In general, higher frequencies required greater intensities for equivalent volume, and the rate of growth in volume (as intensity increased) varied with frequency.

Studies by Fletcher and Munson [1933] yield equal loudness contours that are represented by the Munson-Fletcher curves shown in Figure 2.1. Each point

on any curve represents a particular frequency-intensity information and any point on an equal loudness curve is as loud as any other point on the same curve. The curves represent the auditory behaviour, but they enable loudness comparisons only at the less than, equal to, or greater than level.

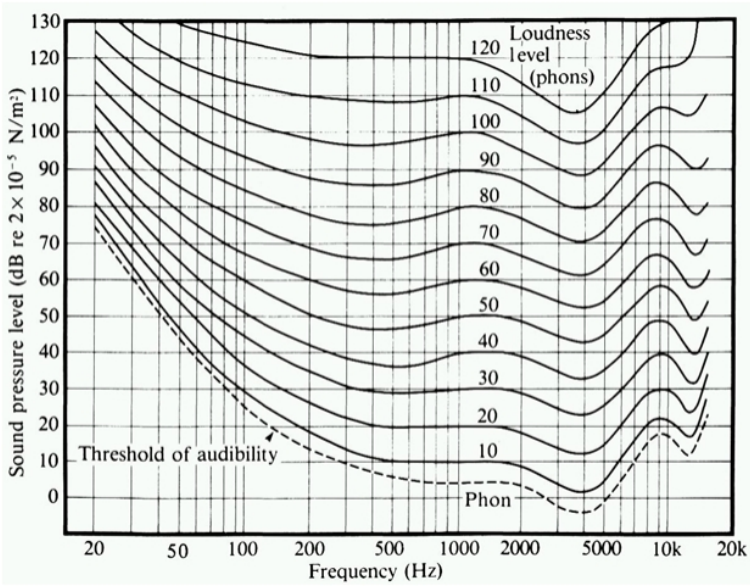

Figure 2.1: Sample equal loudness curves. The curves connect frequency-intensity level combinations which are judged to be equally loud as a 1000Hz standard of a given intensity level ([Radocy and Boyle, 2003, pg.58]). Figure derived from website `https://blogs.msdn.microsoft.com/audiofool/2007/02/07/louder-sounds-better/`, accessed 20th December 2016.

Each curve has a phon value. Phons is a measurement unit for loudness level (not loudness), thus are measures of equivalence to the intensity level of some standard. Phons are arbitrarily set to be equivalent to decibels at 1000 Hz; e.g., a 1000 Hz tone of 60 decibels is also 60 phons. Any other frequency judged as equal in loudness to the 1000 Hz–60 decibels standard is also 60 phons.

2.1.1 Signal properties and its relationship to perceived loudness

Apart from the integration along a perceptual frequency axis as described above, perceived loudness is related to a few other essential properties of the auditory system in either frequency or time domain, as presented by Nielsen and Skovenborg [2004]. The properties that are related to the frequency domain include the phenomenon of masking, that is the affection of the perception of one sound when a louder sound is present at the same time. Also, the phenomenon of spreading, closely related to masking, describes a complex interaction between

different frequencies that are sounded simultaneously: not only the actual frequencies of a signal contribute to the loudness, but also to some extent the higher and lower neighbouring frequencies.

Another property of loudness perception is the spectral loudness summation which is associated with the concept of critical bandwidth; the loudness of a signal within one critical band is independent of its bandwidth, however when the signal's bandwidth grows outside the critical band there is an increase in loudness despite the consistency of the total level.

Applying varying degrees of multi-band dynamics compression affects on loudness which is contrary to the resulting fluctuation depth as Moore [2005] describes in his study to speech signals. Nielsen and Skovenborg [2004] points out that "one of the interesting conclusions was that the long-term loudness, for a given signal, increases with the degree of dynamics compression applied, and with the RMS value held constant", where RMS is the Root Mean Square of the audio signal that is obtained by squaring the amplitude at each instant, acquiring the average of the squared values over the interval of interest, and then taking the square root of this average.

Filtering processes in hearing that influences the loudness perception include the direction dependent filtering, the phenomenon of binaural loudness summation which is the result of having two sensors of sound (ears), and the altering of the sound signal by certain characteristics of the environment, for example the reverberation characteristic.

Properties that are related to the time domain affect the way we perceive loudness. There is a delay between the appearance of an audio signal and the reaction to it by human hearing. Also the post-masking effect is the phenomenon of the perception of a sound fading out, although its source instantaneous stopped producing it. The hearing ability is reduced for a period of time when we are exposed to loud sound for a certain duration. Both the bandwidth and the duration of a burst or tone affect the perceived loudness, however "no simple time constant can be specified for the temporal loudness integration." (Nielsen and Skovenborg [2004])

2.1.2 Sone values as source of loudness information

Laboratory attempts to measure loudness in terms of how much louder one sound is than another have used observers' productions and/or estimations of various loudness distances or ratios. A sone scale of loudness, based on estimates of apparent ratios, and a lambda scale, based on equal-appearing intervals and ratios, are alternative ways of describing apparent loudness relationships (Beck and Shaw [1967]). The reference of the sone scale in particular is that a 1000

Hz sine tone with a level of 40 dB SPL (decibels in reference to sound pressure level) has a loudness of one sone and it is defined by the following equation:

$$\Psi(sones) = \frac{1}{15.849}\left(\frac{I}{I_0}\right)^{0.3}, \tag{2.3}$$

where $I_0 = 10^{-12}\mathrm{W/m}^2$ (Hartmann [2004]).

In our research we have used the sone values at the loudness extraction process from the audio signals. One main reason was the fact that the sone scale is linear, so in this way we can normalise in a more accurate and easy way across different recorded environments. Another equally important reason is that we wanted to train our models on audio that is pre-processed based on the basic psychoacoustic concept of the equal loudness curves, and not to make any other compression or modification. If we want a *machine* to *listen* like a human, then it needs a similar input to the one that a listener has.

For the experiments that are presented in this thesis, the loudness information is extracted from each audio signal using the `ma_sone` function in Elias Pampalk's Music Analysis toolbox[1]. The specific loudness sensation in sones per critical-band is calculated by following the process as explained by Pampalk et al. [2002]:

We calculate the power spectrum of the audio signal using a Fast Fourier Transformation. We use a window size of 256 samples, hopsize of 128, and a Hanning window with 50% overlap. The frequencies are bundled into 20 critical-bands and these frequency bands "reflect characteristics of the human auditory system, in particular of the cochlea in the inner ear" (Pampalk et al. [2002]). Then we calculate the Spectral masking effects, based on the research by Schroeder et al. [1979]. Then we calculate the loudness as dB-SPL unit and from these values we calculate the equal loudness levels in Phon. From these values we detect the values in sones, following the calculation described by Bladon and Lindblom [1981]: if loudness level in phons is L, then it is converted to loudness level in sones S following the formula:

$$S = \begin{cases} 2^{(L-40)/10}, & P \geq 40 \\ (L/40)^{2.642}, & P < 40. \end{cases} \tag{2.4}$$

The extracted loudness time series have been smoothed by local regression using a weighted linear least squares and a 2nd degree polynomial model (the "loess" method of MATLAB's `smooth` function[2]). The loudness time series for each recording is normalised by dividing with its maximum loudness value. In

[1] `www.pampalk.at/ma/documentation.html`, accessed 20 February 2016.

[2] `http://uk.mathworks.com/help/curvefit/smooth.html?refresh=true`, accessed 20 February 2016.

this way we are able to compare different recording environments.

2.1.3 Alternative loudness extraction techniques from music signal

Below we present two techniques that exist and are used for extracting the loudness from a music signal. The first one is called "Power curve"[3] from the plugins that have been created through the Mazurka project[4]. At the synopsis they describe it as measuring the power over time of a signal. They get the average power in decibels for a region of audio data following the equation

$$P_{avg} = 10 \log_{10} \Big(\frac{1}{N} \sum_n x_n^2 \Big), \tag{2.5}$$

where N is the number audio samples to average over, and x_n are the individual audio samples. In addition the plugin can provide a more sophisticated outcome, such as a time-symmetric exponentially smoothed function of the average raw power for each input signal region block, a smoothed power slope of the power curve, as well as a scaled power slope of the smoothed power curve.

The second one is the "intensity" plugin from the BBC audio plugins[5]. It calculates the intensity of a signal following the work by Lu et al. [2006]. As it is described at the plugin documentation, firstly the signal is divided into i subbands with the following frequency ranges:

$$\Big[0, \frac{\omega_0}{2^i}\Big), \Big(\frac{\omega_0}{2^i}, \frac{\omega_0}{2^{i-1}}\Big), \ldots, \Big(\frac{\omega_0}{2^2}, \frac{\omega_0}{2^1}\Big], \tag{2.6}$$

where ω_0 refers to the sampling rate. The intensity of each signal frame n is calculated by:

$$I(n) = \sum_{k=0}^{\omega_0/2} A(n,k),$$

summing the magnitude A of each frequency bin k.

2.1.4 Psychological studies of loudness perception in music

Several studies included in psychoacoustical literature have been attempted to obtain relationships between the subjective magnitude of loudness change and the physical magnitude of intensity change using pure tones or noise (see, for

[3] `http://sv.mazurka.org.uk/MzPowerCurve/`, accessed 18 October 2016
[4] `http://mazurka.org.uk/`, accessed 18 October 2016
[5] `https://github.com/bbc/bbc-vamp-plugins`, accessed 18 October 2016

example Moore et al. [1997] and references therein). However studies using music stimuli in their experiments (see, for example Geringer [1993] and references therein) showed that music stimuli in context were perceived somewhat differently than were the pure tone and noise-band stimuli of previous research.

Following the concept of dynamic changes using music stimuli, the study contacted by Geringer [1995] investigated listener perception of dynamic change within a musical context, regarding both stimulus presentations and subject-response procedures. Musician and non-musicians responded continuously during excerpts using the Continuous Response Digital Interface (CRDI) to indicate perceived loudness levels. Excerpts were selected from different music genres, including instrumental jazz, popular vocal, symphonic orchestra, solo piano, and synthesizer examples. The results revealed that musician subjects indicated a significantly smaller magnitude of dynamic change than non-musician subjects did.

2.2 The concept of score notation

"We call it music, but that is not music:
that is only paper."
—Leopold Stokowski

The concept of music notation, that is the methods of writing down music so that it can be performed, was not involved for a long period in the world's music history. However, it was one of the key points for the development of the European classical music.

An early system of notation was the neumatic notation. Neumes, from the Greek word "neuma", meaning "gesture" or "sigh", was the system of musical notation used from the 7th to 14th century that evolved from representing grave and accute accents to forming a system of a precise indication of pitch for singing (Rutherford-Johnson et al. [2012]). [Cook et al., 2009, pg.11] refers to a conclusion for neumatic notation from a study by Sam Barrett (Barrett [2008]), mentioning that "'music is in other words conceived platonically, as an abstract and enduring entity that is reflected in notation, with the notation itself being reflected in singing (since mistakes in singing can be corrected by reference to the notation)."

During the development of the notation to the form we know today, there were changes mostly on the representation of the duration and the pitch of the notes that are sounded. Innovations included the development of notational symbols for different playing techniques and performance actions. The composer who specified dynamics for the first time in a score was Giovanni